

Offline Signature Forgery Detection with EfficientNet-B0 and Batch-Hard Triplet Loss Fine-Tuning

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Abstract—Handwritten signatures remain a widely accepted biometric trait for authenticating financial, legal, and administrative documents, yet manual forensic verification is slow, subjective, and unreliable against skilled forgeries. This study aims to develop an automated, writer-independent offline signature verification system that generalizes across multiple datasets and that explicitly separates random-impostor from skilled-forgery performance, a distinction often omitted in prior work. Our primary contribution is a lightweight fine-tuning strategy for EfficientNet-B0 that freezes the pretrained convolutional backbone and retrains only its embedding head using batch-hard triplet loss over a combined pool of 221 writers drawn from the CEDAR, GPDS, and a locally collected dataset. For the methodology, signatures were preprocessed, embedded into a 512-dimensional L2-normalized space, and compared by cosine similarity; validation-selected checkpoints were then evaluated on 33 strictly held-out writers. After fine-tuning, random-impostor detection improved from an Area Under the Curve (AUC) of 0.970 to 0.985 (Equal Error Rate reduced from 8.14% to 5.96%), while skilled-forgery detection improved from an AUC of 0.780 to 0.899 (Equal Error Rate reduced from 29.61% to 18.77%). This study demonstrates that targeted, backbone-frozen metric-learning fine-tuning substantially narrows the gap between random-impostor and skilled-forgery detection, establishing a transparent and reproducible benchmark for future cross-dataset offline signature verification.

Keywords—EfficientNet, Metric Learning, Offline Signature Verification, Skilled Forgery Detection, Triplet Loss

I. INTRODUCTION

Handwritten signatures continue to serve as a legally recognized and socially trusted means of authenticating banking transactions, contracts, and official documents, despite the availability of digital authentication alternatives. In practice, the authenticity of a signature is still frequently judged by a human examiner who compares a questioned specimen against known reference samples. This manual process is inherently subjective, difficult to scale, and vulnerable to

examiner fatigue and inter-rater disagreement, particularly when the questioned signature is a *skilled forgery* deliberately practiced to imitate the shape, slant, and stroke rhythm of a genuine writer [1]. These limitations have motivated decades of research into automated offline signature verification (OSV), in which a system must decide, from a static scanned image alone, whether a signature was produced by the claimed writer [1].

The state of the art in OSV has evolved rapidly. Early systems relied on handcrafted image descriptors, such as the Scale-Invariant Feature Transform (SIFT) and the Histogram of Oriented Gradients (HOG), combined with Support Vector Machine (SVM) classifiers; these pipelines were interpretable but generalized poorly across writers because they encoded human-designed notions of similarity rather than learning them from data. The field was subsequently transformed by deep Convolutional Neural Networks (CNNs), with compound-scaled architectures such as EfficientNet demonstrating strong performance while remaining parameter-efficient [2]. The standard practice became transfer learning, in which models pre-trained on large datasets such as ImageNet are fine-tuned for signature verification [3], [4]. More recently, metric-learning formulations based on Siamese and triplet-loss networks have become the de facto standard for writer-independent verification [5], [6], [7], while self-supervised Vision Transformers such as DINOv2 [8] represent the current representation-learning frontier, although their application to signature verification remains largely unexplored.

Despite this progress, a clear research gap remains. Public OSV benchmarks such as CEDAR (55 writers) and GPDS are small relative to other biometric domains, and near-perfect results reported on a single easy-to-saturate dataset, for example an Equal Error Rate (EER) below 0.06% on CEDAR alone [9], do not necessarily transfer to writer-independent, cross-dataset deployment. More critically, skilled forgeries remain substantially harder to reject than random-impostor signatures, yet many studies report only an aggregate accuracy figure without separating these two attack types, obscuring a system's true robustness against a deliberate forger. In addition,

existing CNN transfer-learning approaches typically fine-tune the entire network on a single dataset, which is computationally expensive and offers no evidence of how much of the gain stems from the learned embedding rather than the pretrained backbone.

This study aims to address these gaps through a three-part objective. First, it seeks to build a writer-independent EfficientNet-B0 embedding model evaluated jointly across the CEDAR, GPDS, and a locally collected signature dataset. Second, it aims to design and rigorously evaluate a lightweight, backbone-frozen, batch-hard triplet-loss fine-tuning strategy for the embedding head, isolating its contribution from the pretrained backbone. Third, it aims to separately and transparently quantify performance against random-impostor and skilled-forgery attacks on a strictly writer-disjoint held-out set. This research is justified by the need for accurate, reproducible, and honestly benchmarked signature verification tools, and by the observation that the skilled-forgery scenario, which is the most consequential in practice, is frequently under-reported in the existing literature.

II. LITERATURE REVIEW

A. EfficientNet

EfficientNet is a family of convolutional neural networks introduced by Tan and Le [2], built around a compound scaling method that jointly scales network depth, width, and input resolution using a single coefficient, in contrast to earlier CNNs that scaled these dimensions independently and heuristically. The baseline model, EfficientNet-B0, was obtained through neural architecture search and achieves accuracy competitive with substantially larger networks, such as ResNet-50 and Inception-v3, while using an order of magnitude fewer parameters and floating-point operations. This parameter efficiency makes EfficientNet-B0 particularly attractive for biometric verification, where labeled training data is comparatively scarce and a compact backbone reduces the risk of overfitting when fine-tuned on a few thousand images. EfficientNet-B0 and its variants have consequently been adopted, via transfer learning, in signature verification pipelines that repurpose ImageNet-pretrained features for genuine-versus-forged discrimination [3], [4], typically by replacing the final classification layer with a task-specific head, as is done in this study.

B. Triplet Loss and Metric Learning

Whereas a conventional classifier learns to assign an image to a fixed set of classes, metric learning instead learns an embedding function such that a distance computed between two embeddings reflects semantic identity, a formulation naturally suited to writer-independent verification, where the set of writers at deployment time is unknown during training. The triplet loss, popularized for face verification by Schroff et al. [5], optimizes an embedding by sampling an anchor, a positive example of the same identity, and a negative example of a different identity, penalizing configurations in which the anchor-negative distance is not larger than the anchor-positive distance by at least a margin. Naive triplet sampling, however, produces mostly easy triplets that contribute negligible gradient once basic separation is achieved. Hermans et al. [6] addressed

this with batch-hard mining, which, for every anchor in a batch, selects the hardest positive and the hardest negative actually present in that batch, a strategy adopted directly in this work. In the signature domain, Dey et al. introduced SigNet, a Siamese convolutional network trained with a contrastive objective for writer-independent verification, demonstrating that learned embeddings outperform handcrafted descriptors [7].

C. Related Works

Al-Azzani and Musleh (2024) [3] proposed a Particle Swarm Optimization (PSO)-driven hyperparameter search for a CNN-based offline signature verifier, tuning convolutional filter counts, dense-layer width, dropout rate, and learning rate. The method was evaluated across four datasets, namely GPDS, CEDAR, BHSig260-Bengali, and BHSig260-Hindi, and reported a testing accuracy of 98.3%. The study demonstrates that systematic hyperparameter optimization alone can yield accuracy competitive with more elaborate architectural changes; however, it reports a single aggregate accuracy figure without decomposing performance into random-impostor versus skilled-forgery subsets, limiting direct comparability with forgery-specific robustness claims.

Chokshi et al. (2023) [9] introduced SigScatNet, which couples a Siamese deep network with Scattering wavelet transforms to obtain a computationally lightweight signature embedding suitable for low-resource hardware. Evaluated on the CEDAR dataset and the ICDAR SigComp Dutch dataset, the method achieved an EER of 0.0578% on CEDAR and 3.689% on the SigComp Dutch dataset. While the CEDAR result represents near-perfect separation, the large gap between the two datasets' EER values suggests that CEDAR, with only 55 writers and a fixed forgery set, may be prone to saturation and does not, on its own, stress-test writer-independent generalization, motivating the multi-dataset protocol adopted in this study.

Ozyurt et al. (2024) [4] applied MobileNetV2 as a transfer-learning feature extractor over a dataset of 12,600 signature images from 420 individuals, with 30 genuine signatures per writer, followed by feature selection using Neighborhood Component Analysis (NCA), Chi-squared, and mutual-information criteria before classification with SVM and k-nearest-neighbor classifiers. Feature selection improved accuracy from 91.3% using all extracted features to 97.7% using 300 NCA-selected features, underscoring the value of refining transfer-learned CNN features rather than using them unmodified. Because the described dataset consists exclusively of genuine signatures, however, the reported task is closer to writer identification than to skilled-forgery detection, leaving the harder skilled-forgery scenario, which is the focus of the present work, unaddressed.

III. METHODOLOGY

This section describes the methodology adopted to develop a writer-independent offline signature verification system. The proposed framework follows a structured pipeline, beginning with dataset preparation and preprocessing, proceeding through frozen-backbone feature extraction and embedding-head fine-tuning, and culminating in checkpoint selection and held-out

evaluation under two distinct attack scenarios. Each component is designed to ensure fair, writer-disjoint evaluation and reproducibility. All experiments were implemented in Python using the PyTorch framework and executed on a CPU environment, with pre-computed backbone features cached to disk to keep training tractable.

A. Dataset

For this study, three signature sources were combined: the CEDAR signature database (55 writers, with 24 genuine and 24 skilled-forged signatures per writer), a locally available subset of the GPDS signature database (150 writers, with an average of 8 genuine and 14 skilled-forged signatures per writer), and a locally collected dataset of 16 writers contributing 257 genuine signatures with no forged counterparts. Only genuine signatures were used for embedding fine-tuning, consistent with the standard verification protocol in which a system is enrolled solely on genuine references and encounters forgeries only at test time. The composition of each source is summarized in Table I.

TABLE I. DATASET COMPOSITION

Dataset	Writers	Genuine	Forged	Role in this study
CEDAR	55	1,320	1,320	Fine-tuning (genuine) + forgery test
GPDS (subset)	150	1,200	2,100	Fine-tuning (genuine) + forgery test
Locally collected	16	257	0	Fine-tuning (genuine) only
Combined pool	221	2,052	3,420*	188 train / 33 held-out

*Forged images were never used for training; they were reserved exclusively for the skilled-forgery test in Section III-B and the evaluation in Section IV.

The combined pool of 221 writers was split at the writer level, rather than the image level, into 188 writers for fine-tuning and 33 writers held out entirely for validation and testing, using a fixed random seed to ensure reproducibility and to guarantee that no writer appears in both partitions, a requirement for a valid writer-independent generalization claim. The held-out set comprised 18 GPDS writers, 12 CEDAR writers, and 3 locally collected writers.

B. Proposed Method

The proposed method follows the multi-stage pipeline illustrated in Fig. 1. The workflow begins with preprocessing and feature extraction, proceeds to embedding-head fine-tuning with batch-hard triplet loss, and concludes with checkpoint selection and held-out evaluation.

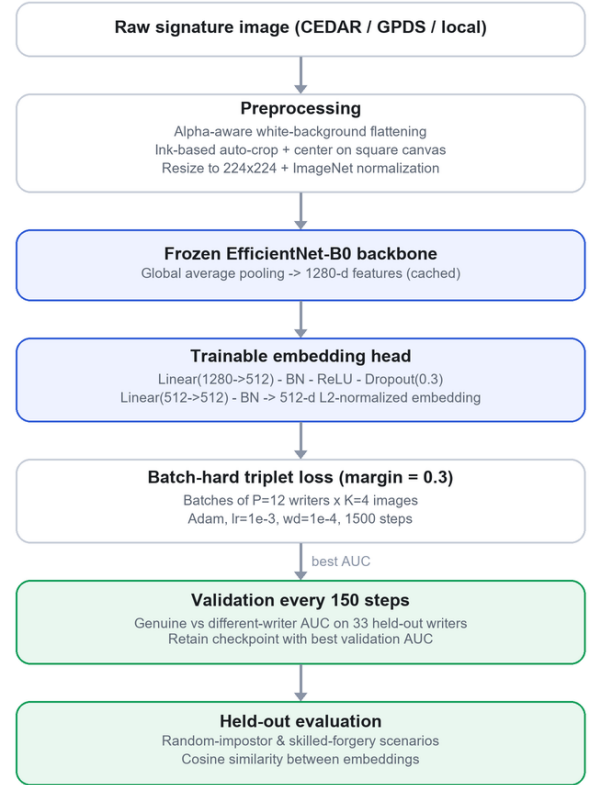


Fig. 1. System flowchart of the proposed pipeline.

During preprocessing, each input image is first converted to a consistent RGB representation: images carrying genuine alpha-channel transparency are composited onto a white background, while fully opaque images are passed through unmodified, avoiding a background-flattening artifact that arises when opaque images are incorrectly treated as transparent. The signature is then auto-cropped to its ink bounding box with a small padding margin, centered on a square canvas, resized to 224×224 pixels, and normalized using the ImageNet channel statistics.

The embedding network consists of an EfficientNet-B0 convolutional backbone, initialized from a checkpoint previously trained end-to-end with triplet loss on GPDS, followed by a projection head composed of a linear layer (1280 to 512), batch normalization, a rectified linear unit, dropout with probability 0.3, a second linear layer (512 to 512), and a final batch-normalization layer, with the output L2-normalized to the unit hypersphere. This pretrained model, used without further modification, constitutes the baseline against which the contribution of this study is measured.

Because the backbone was already trained on a related distribution, it was hypothesized that most of the achievable gain from additional data lay in re-adapting the embedding head rather than the low-level convolutional features. The backbone was therefore frozen entirely, and only the projection head was fine-tuned on the combined 221-writer genuine-only pool. Backbone features of 1,280 dimensions were pre-computed once and cached, enabling efficient training on CPU hardware. Training used batch-hard triplet loss with a margin of 0.3 over cosine distance, with each batch constructed from

twelve randomly sampled writers and four images per writer. The Adam optimizer was used with a learning rate of 1×10^{-3} and a weight decay of 1×10^{-4} for 1,500 steps. Validation AUC on the 33 held-out writers was computed every 150 steps, and the checkpoint with the highest validation AUC was retained as the final proposed model.

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